

Classical Fourth-Order Runge–Kutta

Four-stage explicit integration

1 Butcher tableau and stages

The classical method has tableau

0	0	0	0	0
$\frac{1}{2}$	$\frac{1}{2}$	0	0	0
$\frac{1}{2}$	0	$\frac{1}{2}$	0	0
1	0	0	1	0
	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{6}$

and, for $\dot{z} = f(t, z)$,

$$k_1 = f(t_n, z_n), \quad (1)$$

$$k_2 = f(t_n + h/2, z_n + hk_1/2), \quad (2)$$

$$k_3 = f(t_n + h/2, z_n + hk_2/2), \quad (3)$$

$$k_4 = f(t_n + h, z_n + hk_3), \quad (4)$$

$$z_{n+1} = z_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \quad (5)$$

The non-autonomous stage times are part of the model.

2 Order and stability

The tableau satisfies the Runge–Kutta order conditions through order four; the local error is $\mathcal{O}(h^5)$ and the global error is $\mathcal{O}(h^4)$ for sufficiently smooth problems. On $y' = \lambda y$, its stability polynomial is

$$R(\xi) = 1 + \xi + \frac{\xi^2}{2} + \frac{\xi^3}{6} + \frac{\xi^4}{24}.$$

The finite stability region does not make RK4 an A-stable method.

3 Geometric properties

Classical RK4 is not symmetric and is not symplectic for a general Hamiltonian system. Its high local accuracy can nevertheless make short-time energy and area errors small. The distinction is structural: symplecticity is the exact identity

$$(D\Phi_h)^T \Omega D\Phi_h = \Omega,$$

not merely a small defect at one step size. The implementation exposes each complete step map to observers so local and accumulated defects can be measured.

4 Energy-tracking configuration

The public model has an optional `track_energy` configuration. When enabled for dynamics satisfying the extended Hamiltonian contract, RK4 advances the triangular state (z, t, κ) with

$$\dot{k} = -\partial_t H(t, z), \quad K(z, t, k) = H(t, z) + k.$$

The same RK4 nodes and weights are used for z and the normalized momentum. The physical trajectory is returned normally; k and the maximum sampled drift of K are diagnostics. This extension monitors generalized energy but does not turn RK4 into a symplectic or energy-preserving method.

5 Implementation correspondence

The method is implemented in `src/methods/classical/rk4.py` and consumes the generic dynamics protocol. Main-grid steps are independent of the output schedule; shadow steps provide off-grid samples without changing the accepted trajectory or emitting observations. See the sibling runtime document for the exact result and diagnostic contract.

6 Four state formulations and passive energy tracking

For one planar particle the supported internal representations are

physical	$z = (x, y)$	$\mathbb{R}^2$,
physical with energy	(z, t, κ)	$\mathbb{R}^4$,
duplicated	$(u, v), \quad u, v \in \mathbb{R}^2$	$\mathbb{R}^4$,
duplicated with energy	(u, v, t, κ)	$\mathbb{R}^6$.

Classical methods use the physical rows; ABBA and BM4 use the duplicated rows. `track_energy` selects the energy variant. For N planar particles the respective dimensions are $2N$, $4N$, $4N$, and $6N$: one time entry and one momentum per particle. All time entries equal the integration time. Classical full-cyclotron runs instead use $4N$ or $6N$ entries. Accepted spatial copies coincide; only one physical copy is exported. Every coordinate block contains N values in component-major order. The formulation validates particle times and aligns output round-off; the integrator does not interpret coordinate indices.

The normalized passive balance is

$$\dot{\kappa} = -\partial_t H(t, z), \quad \kappa(t_0) = 0, \quad \Delta K(t) = H(t, z(t)) + \kappa(t) - H(t_0, z_0).$$

Splitting accumulators from both copies are divided by two. The physical Hamiltonian may vary with time; ΔK measures its computed balance. Tracking leaves physical trajectories, nonlinear stopping rules and adaptive acceptance unchanged. Dynamics must supply the momentum derivative, including zero for an autonomous Hamiltonian.

Hairer's constraint is solely $u - v = 0$: the reduced multiplier has $2N$ entries and ABBA's simultaneous spatial solve has $6N$ unknowns. Neither t nor κ is projected. The former eight-component projection is retired; `state_extension='fully_extended'` raises migration guidance.

Dynamics, formulation, method and integration remain separate. The shared state classes own layouts, diagonal embedding and history extraction in

`src/formulations/state.py`

Each integration initializes a fresh method instance. Observers receive physical states; energy histories are in `Solution.diagnostics`. The `extended_time` history has shape (N, n_{samples}) . Fixed-step shadow samples leave main states and counters unchanged. Adaptive energy quadrature uses accepted dense output; its backend state and Jacobian remain physical.

The passive momentum uses the same four stage states and weights as the physical RK4 update.

7 CPU and JAX execution of the same map

The execution configuration selects either the default NumPy/SciPy CPU driver or a JAX driver on one CPU or GPU device. Both consume the same RK4 stage and passive-quadrature formulas. The potential uses the same fitted spline knots and coefficients in both implementations. JAX requires explicit double precision; different floating-point operation ordering can produce round-off differences, so equivalence does not mean bitwise identity.

For independent particles, stage operations are batched over the particle index. The four stages retain their dependencies, and successive time steps remain sequential inside a compiled loop. The accepted state and requested sample buffer remain on the chosen device until the completed result is transferred to the standard immutable NumPy solution. Independent shortened maps still provide off-grid samples without altering accepted main states. The passive momentum and sampled Hamiltonian are evaluated on that device.

The compiled route currently covers the built-in guiding-centre and full-cyclotron equations. Custom dynamics, method subclasses and Python step callbacks or progress reporting require the CPU route. Device choice does not change the order, stability region or geometric properties derived above. First-call compilation must be distinguished from warmed integration timings, which include transfers and output construction in the test study.